

Python Programs

1. Ternary Operator — 5 Questions

1. *Positive or Negative*

```
num = int(input("Enter a number: "))

result = "Positive" if num >= 0 else "Negative"

print(result)
```

2. *Even or Odd*

```
num = int(input("Enter an integer: "))

result = "Even" if num % 2 == 0 else "Odd"

print(result)
```

3. *Eligible to Vote*

```
age = int(input("Enter your age: "))

result = "Eligible to vote" if age >= 18 else "Not eligible"

print(result)
```

4. *Find the Larger Number*

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

larger = a if a > b else b

print("Larger number:", larger)
```

5. *Pass or Fail*

```
marks = int(input("Enter marks: "))

result = "Pass" if marks >= 35 else "Fail"

print(result)
```

2. Match Case — 5 Questions

1. Day of the Week

```
day = int(input("Enter a number (1-7): "))

match day:
    case 1:
        print("Monday")
    case 2:
        print("Tuesday")
    case 3:
        print("Wednesday")
    case 4:
        print("Thursday")
    case 5:
        print("Friday")
    case 6:
        print("Saturday")
    case 7:
        print("Sunday")
    case _:
        print("Invalid day")
```

2. Simple Calculator

```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
operator = input("Enter operator (+, -, *, /): ")

match operator:
    case "+":
        print(a + b)
    case "-":
        print(a - b)
    case "*":
        print(a * b)
    case "/":
        print(a / b)
    case _:
        print("Invalid operator")
```

3. Menu Selection

```
print("1. Add Student")
print("2. View Student")
print("3. Update Student")
print("4. Delete Student")

choice = int(input("Enter your choice: "))

match choice:
    case 1:
        print("Add Student")
    case 2:
        print("View Student")
    case 3:
        print("Update Student")
    case 4:
        print("Delete Student")
    case _:
        print("Invalid choice")
```

4. Grade Description

```
grade = input("Enter grade (A, B, C, D, F): ")

match grade:
```

```
case "A":
    print("Excellent")
case "B":
    print("Very Good")
case "C":
    print("Good")
case "D":
    print("Needs Improvement")
case "F":
    print("Fail")
case _:
    print("Invalid grade")
```

5. Traffic Signal

```
color = input("Enter traffic light color: ")
```

```
match color:
    case "red":
        print("Stop")
    case "yellow":
        print("Wait")
    case "green":
        print("Go")
    case _:
        print("Invalid signal")
```

Note: match-case requires Python 3.10 or later.